

Cloud Physics, Fall 2022 Homework I

BI0209015 劉韋杰

$P_0 = 1015 \text{ hPa}$	$P_1 = 950 \text{ hPa}$	$P_2 = 800 \text{ hPa}$
$T_0 = 31^\circ\text{C}$	$T_1 = 24.880^\circ\text{C}$	$T_2 = 15.895^\circ\text{C}$
$T_d = 22^\circ\text{C}$	$w_{s1} = 16.892 \text{ g/kg}$	$w_{s2} = 14.484 \text{ g/kg}$
	$\chi_1 = 0 \text{ g/kg}$	$\chi_2 = 2.408 \text{ g/kg}$

```

Constant = 54.492616144439346
T_parcel at 950 hPa by potential temperature = 25.30001582074277
T_parcel at 800 hPa = [15.89541642] °C
Total water mixing ratio at 800 hPa = 0.016892493215274317 kg/kg
Vapor mixing ratio at 800 hPa = [0.01448446] kg/kg
Condensed water mixing ratio at 800 hPa = [0.00240803] kg/kg
T_parcel = [24.88012126] °C
Total water mixing ratio at 950 hPa = 0.016892493215274317 kg/kg
Saturated vapor mixing ratio at 950 hPa = [0.02171789] kg/kg
Condensed water mixing ratio at 950 hPa = [0] kg/kg
    
```

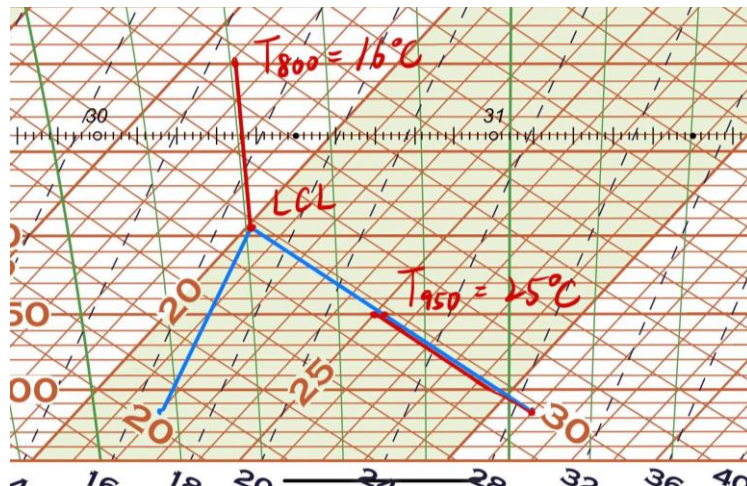
1、從斜溫圖可概略得知，此氣塊飽和處約在 900 hPa 處，凝結溫度約 20 度，950 hPa 處溫度約 25 度，800 hPa 處溫度約 16 度，此可作為結果的驗證。

2、將地面氣壓與溫度代入位溫公式，可得位溫，由於在乾絕熱遞減時位溫守恆，因此將此位溫與 950 hPa 代入可解出溫度為 25.3°C

3、將露點溫度($T=22^\circ\text{C}$)代入 Clausius-Clapeyron relation，可得地面的水氣壓，依此水氣壓和地面氣壓($P=1015 \text{ hPa}$)帶入水氣混和比的估計式，可得 mixing ratio。

在未飽和前的氣塊，水氣量不變 $Q = w_s = w$ 。而飽和後氣塊，水氣量 Q 依然相同，不過 w_s 會隨溫度變化，將 T 代入式(3)得飽和後 w_s 。

代入可逆飽和絕熱公式可得常數值，再將 950、800 hPa、 Q 、 w_s 代入公式，利用 Python 解方程工具可得 950 和 800 hPa 時的溫度



```

import numpy as np
from scipy.optimize import fsolve

# set up parameters

Rd = 287
Cp = 1004
Lv = 2.5e6
Rv = 461
cw = 4187
epsilon = 0.622
e0 = 6.11

# %%

# Clausius-Clapeyron equation
e_1015 = e0 * np.exp( (Lv/Rv) * ( 1/273.15 - 1/(22+273.15) ) )

# w = mv/md = (epsilon * e) / (P - e)
w_1015 = (epsilon*e_1015) / (1015-e_1015)

# Before parcel saturated, X = 0, Q = ws = w (Actual vapor)
# calculate the constant of reversible saturated adiabatic process equation

Q = w_1015
ws = Q
constant =
((31+273.15)/1015**(Rd/(Cp+Q*cw)))*np.exp((ws*Lv)/((31+273.15)*(Cp
+Q*cw)))
print('Constant = ',constant)

# %% potential temperature

P0 = 1000
theta = (31+273.15)*(P0/1015)**(Rd/Cp)
T_950 = theta / ((P0/950)**(Rd/Cp)) - 273.15
print('T_parcel at 950 hPa by potential temperature = ',T_950)

```

#%% use 可逆飽和絕熱公式 to solve out Tc at 800 hPa

T = 273.15

def func2(T):

 P = 800

 return

 (((T/P**(Rd/(Cp+Q*cw))*np.exp((((epsilon*(e0*np.exp((Lv/Rv)*((1/273.15)
- (1/T)))))/(P-e0*np.exp(Lv/Rv*(1/273.15-1/T)))*Lv)/(T*(Cp+Q*cw)))) -
constant)

T_800 = fsolve(func2,T)

print('T_parcel at 800 hPa = ',T_800-273.15 , '°C')

es_800 = e0*np.exp(Lv/Rv*(1/273.15-1/T_800))

ws_800 = epsilon*es_800/(800-es_800)

X_800 = Q-ws_800

print('Total water mixing ratio at 800 hPa = ',Q,'kg/kg')

print('Vapor mixing ratio at 800 hPa = ',ws_800,'kg/kg')

print('Condensed water mixing ratio at 800 hPa = ',X_800,'kg/kg')

#%% use 可逆飽和絕熱公式 to solve out Tc at 950 hPa

T = 300

def func1(T):

 P = 950

 return (T/P**(Rd/(Cp+Q*cw))*np.exp((w_1015*Lv)/(T*(Cp+Q*cw))) -

constant

T_950 = fsolve(func1,T)

print('T_parcel = ',T_950-273.15 , '°C')

es_950 = e0*np.exp(Lv/Rv*(1/273.15-1/T_950))

ws_950 = epsilon*es_950/(950-es_950)

X_950 = Q-ws_950

print('Total water mixing ratio at 950 hPa = ',Q,'kg/kg')

print('Saturated vapor mixing ratio at 950 hPa = ',ws_950,'kg/kg')

```
print('Condensed water mixing ratio at 950 hPa = [0] kg/kg')
```